Math Table: The Eckmann-Hilton Argument

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Hi everyone, my name is Vincent, I'm a junior in Lowell House studying math and physics, and I will be giving today's Math Table talk on the Eckmann-Hilton argument. Before I start, I want to thank the organizers of Math Table, Eva, Hannah and Roderic. I also want to thank Thomas Brazelton and my entryway tutor Francisco for helping me practice my talk. Finally, I want to thank my DRP mentor Matthew Niemiro for introducing me to the Eckmann-Hilton argument, and for being a great friend/mentor. The Eckmann-Hilton argument is a neat trick to show an operation is commutative, which has applications in different areas of math. In today's talk, we will discuss an application in algebraic topology; we will use the argument to prove that the higher homotopy groups are abelian.

Theorem 1 (Eckmann-Hilton)**.** *Let M be a set equipped with two monoid structures $(M, *)$ and $(M, \cdot)$ sharing the same unit $1 \in M$. For those who are unfamiliar with the definition of a monoid, this just means we have two associative operations $*, \cdot : M \times M \to M$ sharing the same identity $1 \in M$. In case this helps with someone's understanding, a monoid is just a group without the requirement that every element has an inverse. The Eckmann-Hilton theorem says that if the two operations satisfy the interchange law*

$$(a \cdot b) * (c \cdot d) = (a * c) \cdot (b * d), \quad \forall a, b, c, d \in M,$$

then they are the same, $a * b = a \cdot b$, and in particular, they are commutative, $a * b = b * a$, for all $a, b \in M$.

Therefore, we can see why the Eckmann-Hilton argument might be useful. If you have some operation and you want prove that it is commutative, you can of course show this directly by computing $a \cdot b = b \cdot a$ for every a, b, but in cases where this is difficult, it might be easier to find another "compatible" operation for which the interchange is satisfied, so you can use the Eckmann-Hilton argument to say they are commutative. We will see an example of this when we consider the higher homotopy groups. For now though, we should prove the theorem.

Proof: Let $a, b \in M$, then

$$a * b = (a \cdot 1) * (1 \cdot b) = a \cdot b,$$

so the multiplications are the same, and

$$a * b = a \cdot b = (1 * a) \cdot (b * 1) = (1 \cdot b) * (a \cdot 1) = b * a,$$

so they are commutative. □

Remark 2. Note that the theorem is actually more powerful than the version we stated. We don't need to assume that both operations have the same identity, nor do we need to assume that they are associative.

The full version of the theorem says that if two unital operations satisfy the interchange law, then they are the same, they have the same identity, they are associative, and they are commutative.

We will now apply the Eckmann-Hilton argument to prove that the higher homotopy groups are abelian. First, let's recall the definition of a homotopy group, starting with the fundamental group.

Definition 3 (Fundamental group)**.** Let X be a topological space and $x_0 \in X$ be a base point. A loop in X based at x_0 is a continuous map $\gamma : [0,1] \to X$ such that the start and end points are equal to the base point, that is, $\gamma(0) = \gamma(1) = x_0$. We can visualize this as (insert sketch of a loop). As a set, the fundamental group of X relative to the base point x_0, which we denote $\pi_1(X, x_0)$, is defined to be the set of loops based at x_0 modulo homotopy. For those who are unfamiliar with the concept of homotopy, what this means is that instead of considering the set of all loops, we define an equivalence relation in which two loops $\gamma_0 \sim \gamma_1$ are equivalent if we can "continuously deform one onto the other" (insert sketch of a path homotopy). We have described $\pi_1(X, x_0)$ as a set, but it is a group, so what is the operation? The operation is concatenation of loops, and this is the picture. Concretely, given by two classes $[\gamma_0], [\gamma_1] \in \pi_1(X, x_0)$ represented by loops $\gamma_0, \gamma_1 : [0,1] \to X$, their product $\gamma_0 \cdot \gamma_1 : [0,1] \to X$ is defined as the loop

$$(\gamma_0 \cdot \gamma_1)(t) = \begin{cases} \gamma_0(2t), & 0 \leq t \leq \frac{1}{2} \\ \gamma_1(2t-1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Note that this just means "follow γ_0 first with twice the speed, and then follows γ_1 with twice the speed". We then define the product of classes as the product of their representative loops $[\gamma_0] \cdot [\gamma_1] = [\gamma_0 \cdot \gamma_1]$.

Remark 4. The fundamental group is not abelian, so we will have nothing more to say about it for now. The definition was introduced to motivate the definition of the higher homotopy groups.

Definition 5 (Higher homotopy groups)**.** Let X be a space and $x_0 \in X$ a base point. The homotopy group $\pi_n(X, x_0)$ is defined as the set of homotopy classes of continuous maps $\gamma : [0,1]^n \to X$ such that the boundary $\partial[0,1]^n$ is sent to the base point, that is, $\gamma(\partial[0,1]^n) = x_0$, and the operation is concatenation of cubes, glued together along some face. Concretely, given two classes $[\gamma_0], [\gamma_1] \in \pi_n(X, x_0)$ represented by maps $\gamma_0, \gamma_1 : [0,1]^n \to X$, their product can be defined as the map

$$(\gamma_0 \cdot \gamma_1)(t_1, t_2, \cdots, t_n) = \begin{cases} \gamma_0(2t_1, t_2, \cdots, t_n), & 0 \leq t_1 \leq \frac{1}{2} \\ \gamma_1(2t_1 - 1, t_2, \cdots, s_n), & \frac{1}{2} \leq t_2 \leq 1. \end{cases}$$

Note that this concatenates the cubes along the first coordinate in the product $[0,1] \times [0,1] \times \cdots \times [0,1]$, following γ_0 with twice the speed along the first coordinate, and then γ_1 with twice the speed. The picture for $n = 2$ is the following. We then define the product of classes once again as $[\gamma_0] \cdot [\gamma_1] = [\gamma_0 \cdot \gamma_1]$.

Remark 6. We can see that we recover the fundamental group if we set $n = 1$.

However, for $n > 1$, there is no reason why can't concatenate the cubes along a different coordinate, for example the second one, and define the product instead as

$$(\gamma_0 * \gamma_1)(t_1, t_2, \cdots, t_n) = \begin{cases} \gamma_0(t_1, 2t_2, \cdots, t_n), & 0 \leq t_2 \leq \frac{1}{2} \\ \gamma_1(t_1, 2t_2 - 1, \cdots, s_n), & \frac{1}{2} \leq t_2 \leq 1. \end{cases}$$

Once again, the picture for $n = 2$ is the following. Now, recall that our goal is to show $\pi_n(X, x_0)$ is abelian, that is, the operation is commutative. Since we have defined two operations, and our intuition says they should be the same, some light bulbs are going off in our heads about how we can use the Eckmann-Hilton argument. If we can show that these two definitions of the product satisfy the interchange law, then by the Eckmann-Hilton argument they are in fact the same as we expect, and in particular, they are commutative, which proves that the homotopy groups are abelian for $n > 1$.

Proposition 7. *The products $\cdot$ and $*$ satisfy the interchange law $(\gamma_0 \cdot \gamma_1) * (\gamma_2 \cdot \gamma_3) = (\gamma_0 * \gamma_2) * (\gamma_1 * \gamma_3)$.*

Proof: This follows directly from plugging in the definitions, and I won't go through the details today. However, we will consider the following visual proof sketch for $\pi_2(X, x_0)$. (insert visual proof sketch) □

Corollary 8. *By the Eckmann-Hilton argument, higher homotopy groups $\pi_n(X, x_0)$ for $n > 1$ are abelian.*

I will spend the last few minutes discussing a reformulation of the Eckmann-Hilton argument in the language of category theory. The key point for this reformulation is that the interchange law is equivalent to saying that $* : (M, \cdot)^2 \to (M, \cdot)$ is a morphism of monoids, or equivalently, that $\cdot : (M, *)^2 \to (M, *)$ is a morphism of monoids. This gives the following reformulation of the Eckmann-Hilton argument:

Theorem 9. *A monoid object in the category of monoids is a commutative monoid, that is, Mon(Mon($\mathcal{C}$)) is contained in CMon($\mathcal{C}$).*

Let's unpack what this means. Let $(C, \otimes)$ be a monoidal category, so we can actually consider monoid objects in $\mathcal{C}$. Then, the category $\mathrm{Mon}(\mathcal{C})$ is the category of monoid objects in $\mathcal{C}$. Its objects are triples $(M, \cdot, \eta)$, where M is an object in $\mathcal{C}$, and $\cdot : M^2 \to M$ and $\eta : 1_{\mathcal{C}}$ are morphisms such that the axioms of a monoid are satisfied, that is, $\cdot$ is associative and unital. We want to consider monoid objects in $\mathrm{Mon}(\mathcal{C})$, so we need to be able to take tensor products in $\mathrm{Mon}(\mathcal{C})$. In order to do this, we require that $\mathcal{C}$ be symmetric monoidal, so that if we have two monoids $(M, \cdot)$ and $(M', \cdot')$, we can define their tensor product as $(M \otimes M', \tilde{\cdot})$, where $\tilde{\cdot}$ is defined as

$$(M \otimes M') \otimes (M \otimes M') = M^2 \otimes M'^2 \xrightarrow{\cdot \otimes \cdot'} M \otimes M'.$$

Now, consider a monoid in $\mathrm{Mon}(\mathcal{C})$, which means we have a triple $((M, \cdot), *, \eta)$ where $* : (M \otimes M, \cdot \otimes \cdot) \to (M, \cdot)$ is associative and unital. Since it is a morphism of monoids, this is equivalent to the interchange law being satisfied for $\cdot$ and $*$, it turns out that $\cdot$ is commutative, so $(M, \cdot)$ is in $\mathrm{CMon}(\mathcal{C})$.

Theorem 10. *Let $(C, \otimes)$ be a symmetric monoidal category. Then the category of monoids objects in Mon($\mathcal{C}$) is equivalent to the category CMon of commutative monoids in $\mathcal{C}$.*